

Casino Model

Let M_t denote the money in hand on the t th day and suppose that

$$M_{t+1} = p * M_t * (1 + f) + (1 - p) * M_t * (1 - f)$$

where p is the gambling winning rate and f is the proportion of money to bet on each stage.

Suppose now that the casino has the probability δ to open and the probability $1 - \delta$ to close. When the casino is open, the player can continue on gambling. If casino closes, the player takes the money and then goes home.

If we let $V_t(m)$ denote the maximal expected profit when we have m amount of money and the option is the fraction we choose to bet for every stage.

$$V_t(M_t) = \max_f (1 - \delta) * u(E(\text{Revenue at period } t)) + \delta * E(V_{t-1}(M_{t-1}))$$

where $E(\text{Revenue at period } t) = p * M_t * (1 + f) + (1 - p) * M_t * (1 - f)$

$$E(V_{t-1}(M_{t-1})) = p * V_{t-1}(M_t * (1 + f)) + (1 - p) * V_{t-1}(M_t * (1 - f))$$

and $u(\cdot)$ is the function we would vary to see different outcomes.

Now we plot the relationship between optimal betting proportion f and gamble winning probability p given different casino opening probability δ under different utility function $u(\cdot)$. We can see as $\delta \rightarrow 1$, the result is converged to Kelly criterion result regardless of what the function is. Meanwhile, when $u(x) = x^{1-\delta}$, as $\delta \rightarrow 1$ the result is exactly the same with Kelly criterion result.

Case1: when $u(x) = x^{1-\delta}$

i) Given the casino opening probability $\delta = 0.7$, we plot how optimal proportion f to bet changing with the change in gamble winning probability p .

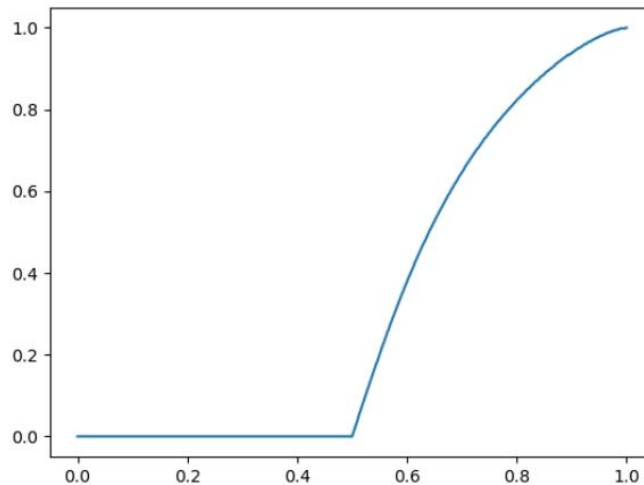
Figure 1 notes:

X-axis: gamble winning prob p

Y-axis: optimal betting fraction f

Conclusion: when $p > 1/2$, that is, when we have positive expected profit for each period, the proportion to bet is positive and increasing concavely in gamble winning probability p .

Figure 1



ii) Given the casino opening probability $\delta = 0.9$ and $\delta = 0.99$, we plot how optimal proportion f to bet changing with the change in gamble winning probability p .

Figure 2 notes:

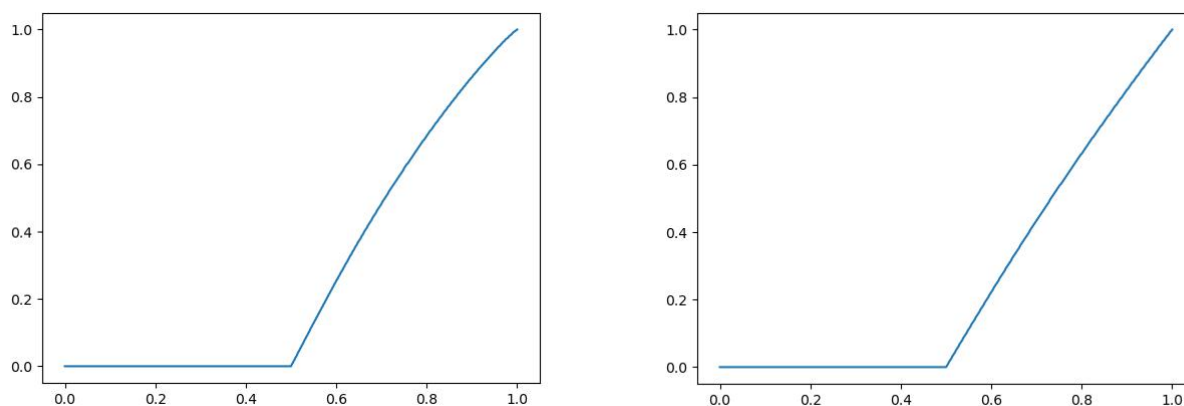
X-axis: gamble winning prob p

Y-axis: optimal betting fraction f

Conclusion: On the left-hand side is when $\delta = 0.9$ and on the right-hand side is when $\delta = 0.99$, we can see when δ is close enough to 1, the result is exactly the Kelly criterion result which is a linear straight line in winning probability p .

To refresh, the Kelly criterion result is the optimal betting fraction $f = p - q = 2p - 1$.

Figure 2



iii) Changing the initial money before starting to bet.

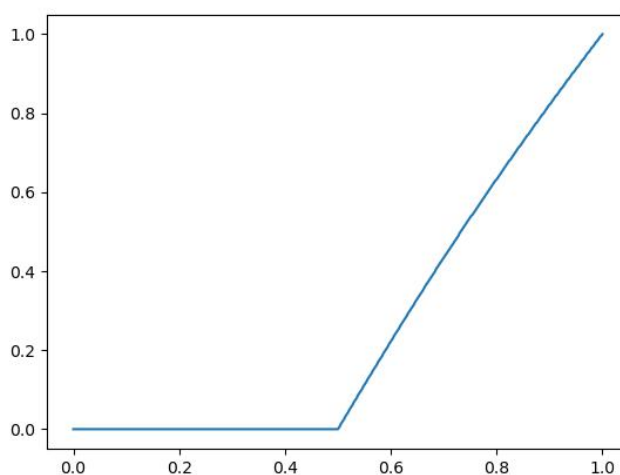
Figure 3 notes:

X-axis: gamble winning prob p

Y-axis: optimal betting fraction f

Conclusion: In (i) and (ii), we set the initial money to 100. To see if the result is robust, we change the initial money to 10000 and it turns out that the result is the same. Especially when $\delta = 0.99$, the result is still the same with the Kelly criterion result.

Figure 3



Case2: when $u(x) = \log(x)$, we set $\delta = 0.7, 0.9, 0.99$ to see the relationship between optimal betting fraction and gamble winning probability.

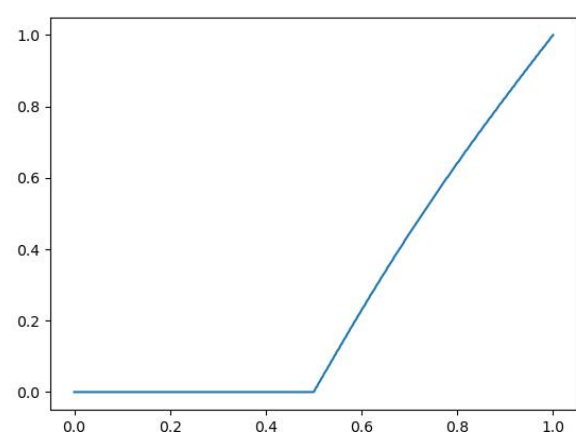
Figure 4 notes:

X-axis: gamble winning prob p

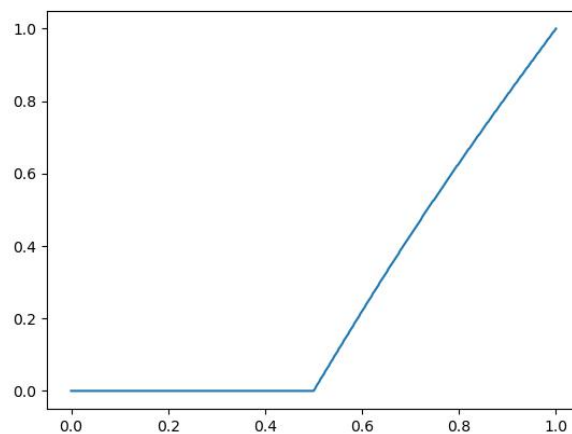
Y-axis: optimal betting fraction f

Conclusion: We can see that under $u(x) = \log(x)$, the result converges faster than when in case where $u(x) = x^{1-\delta}$. However, even when $\delta = 0.99$, the result is not exactly the same but very close to Kelly criterion result under all gamble winning prob p . Though it's not obvious to see, the curve lines are all strictly concave when $p > 1/2$.

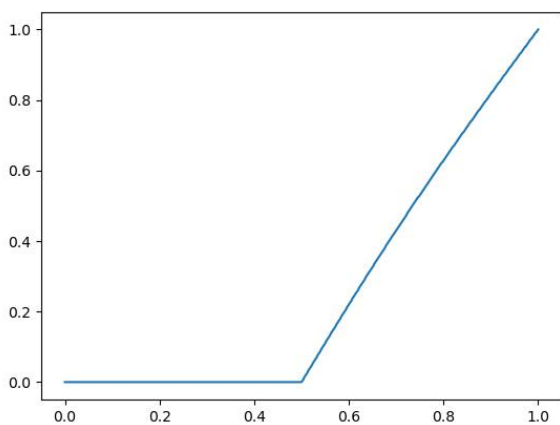
Figure 4



When $\delta = 0.7$



When $\delta = 0.9$



When $\delta = 0.99$

Now we plot the relationship between optimal betting proportion f and casino opening probability δ given different gamble winning probability p under different utility function $u(\cdot)$. To see if the final result is close to Kelly criterion result, we can just see from the last point where δ changing close enough to 1 if the optimal betting fraction is $2 * p - 1$.

Case 1: when $u(x) = x^{1-\delta}$ and gamble winning probability $p = 0.6$.

(i) The initial money= 100.

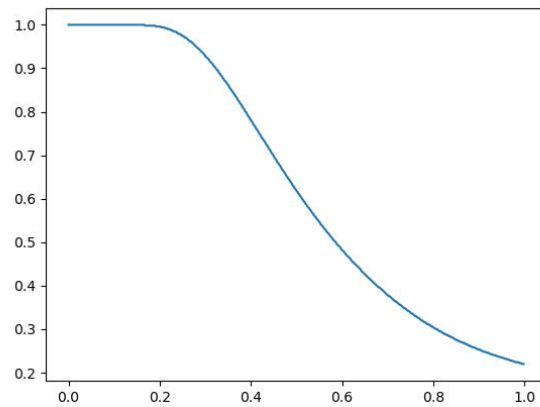
Figure 5 notes:

X-axis: prob of casino is open δ

Y-axis: optimal betting fraction f

Conclusion: The betting fraction drops convexly in δ . There are two points we can see. Firstly, when δ is close enough to 0, that is, when in each period we know this is very likely to be our last chance to gamble, we will go all in and put all of our money into gamble which is intuitive. Secondly, when δ is close enough to 1, the result is the same with Kelly criterion result= $2 * p - 1 = 0.2$.

Figure 5



(ii) To see if the result is robust, we set the initial money= 10000.

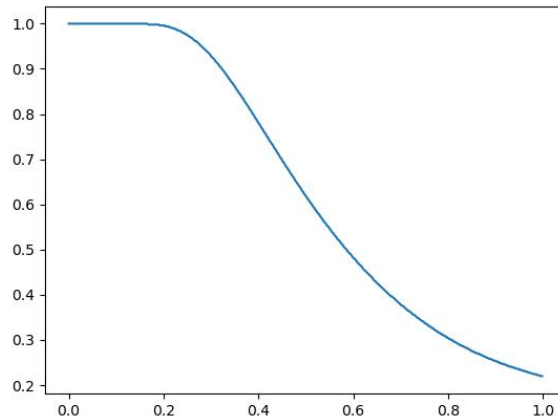
Figure 6 notes:

X-axis: prob of casino is open δ

Y-axis: optimal betting fraction f

Conclusion: The initial money does not affect the result. The result is robust.

Figure 6



Case2: when $u(x) = \log(x)$ and gamble winning probability $p = 0.6$.

Figure 7 notes:

X-axis: prob of casino is open δ

Y-axis: optimal betting fraction f

Conclusion: We can see the extreme cases where $\delta = 1$ and $\delta = 0$ are the same with the result with $u(x) = x^{1-\delta}$. However, the optimal betting fraction in this case drops more smoothly and convexly.

Figure 7

